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AI for Bharat Hackathon

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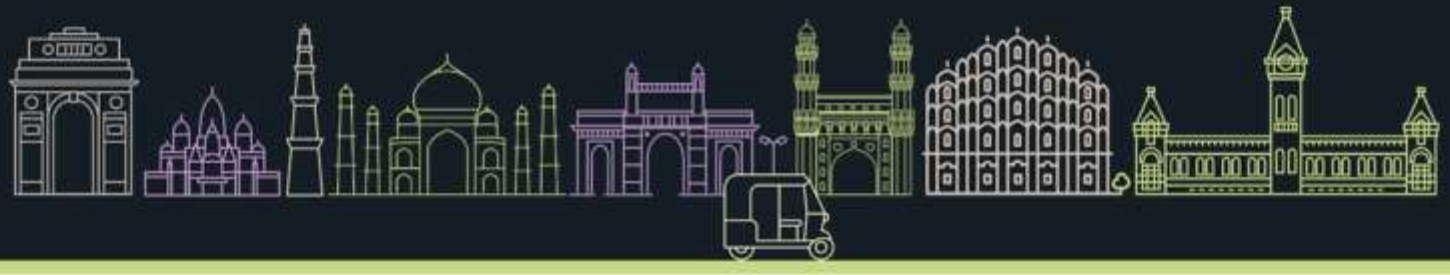

Team Name :Starters

Team Leader Name :kowsik.s

Problem Statement :Missing Persons Rapid Intelligence System

Brief about the Idea:

Thousands of missing children and elderly cases every year. Police depend on manual FIR matching. Posters+WhatsApp forwards are ineffective . Rural areas lack centralized tracking. CCTV footage id rarely analyzed effectively. Keeping in mind these problem statements, we came up with an idea of using Ai to overcome this issue so that we can contribute something effectively to the society .The rough idea is , it accepts missing petrons details (Photo + description). It continuously scans CCTV footages , Railway stations, Bus stands , shelter homes, Public surveillance. It matches faces+clothing+gaint patterns and alerts nearby police stations automatically .The thing which makes it unique is it uses Multi-Modal Ai (Face recognition, cloth color detection, height estimation, walk pattern(gait recognition), Age progression Ai (If the person is missing for long time).

Your solution should be able to explain the following:

- **How different is it from any of the other existing ideas?**
- The thing which makes it unique is it uses Multi-Modal Ai (Face recognition, cloth color detection, height estimation, walk pattern(gait recognition), Age progression Ai (If the person is missing for long time).

- **How will it be able to solve the problem?**

- Age Progression Model (If child is missing for 5 years the AI predicts current face)
- Cloth –Re-Identification Model if face is hidden , it detects based on shirt coloe, backpack, etc
- Gait Recognition : Identify person based on walking style.
- Geo-intelligence alert system- sends alert to nearby police station, generated heatmap of sightings
- **USP of the proposed solution**

- **Multi-Modal AI Intelligence**

Unlike existing systems that rely only on face recognition, our solution integrates face recognition, clothing re-identification, gait analysis, and age progression modeling to increase detection accuracy even when facial visibility is low.

- **Age Progression Prediction Model**

The system uses deep generative AI to predict how a missing child or person may look after several years, improving long-term recovery chances.

- **Real-Time Geo-Intelligent Alerting**

Upon detecting a potential match, the system automatically sends alerts to the nearest police station and generates location heatmaps for faster intervention.

- **Works Even with Partial Visibility**

If the face is masked or unclear, the system can identify individuals using clothing color, accessories, body posture, and walking style.

- **Designed Specifically for Indian Infrastructure**

Optimized for railway stations, bus depots, and crowded public places common in India.

List of features offered by the solution:

1. Multi-Modal Person Identification

Face recognition using deep learning models
Clothing color and accessory detection
Height and body posture estimation
Gait (walking pattern) recognition

2. AI-Based Age Progression Prediction

Predicts how a missing person may look after several years
Useful for long-term missing cases
Uses generative AI models for facial transformation

- **3. Real-Time CCTV Surveillance Integration**
- Continuous monitoring of CCTV feeds
- Automatic detection and matching
- Works in railway stations, bus stands, malls, and public spaces
-

- 📍 **4. Geo-Intelligent Alert System**
- Sends alerts to nearest police station
- Location-based heatmap of possible sightings
- Real-time SMS/email notification to registered guardians
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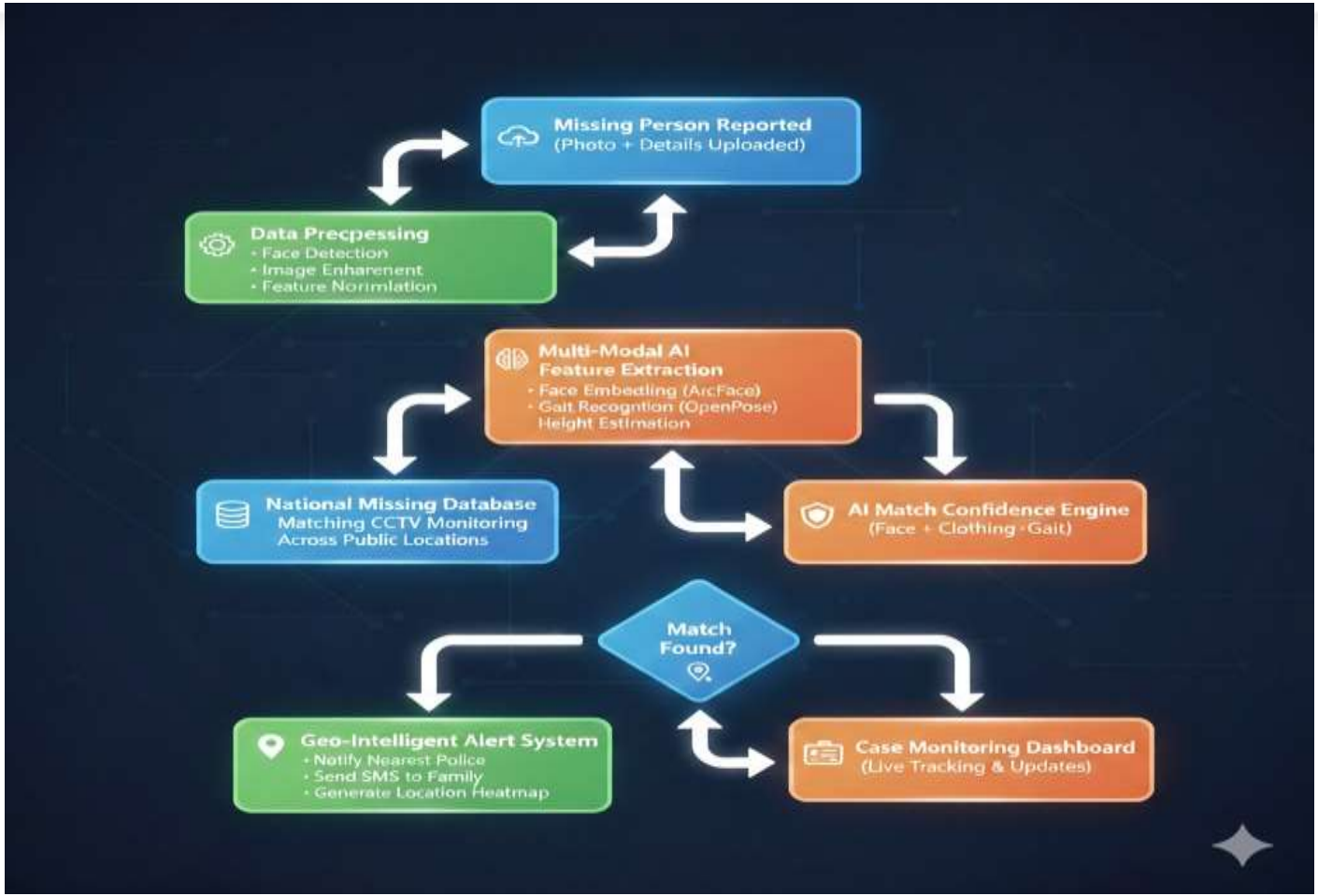
- 🏛 **5. Centralized National Database**
- Stores missing person profiles securely
- Enables cross-state matching
- Integrates with police FIR systems

6. Privacy & Security Framework
Encrypted biometric data
Role-based access for law enforcement
Ethical AI compliance

📱 7. Mobile Application for Police & Authorities
Upload missing person details
Receive instant alerts
View live detection results

🔊 8. Audio-Based Distress Detection (Optional Advanced Feature)
Detects child cry patterns
Recognizes distress keywords like “help”
Supports noisy public environments


Missing Person Reported
(Photo + Details Uploaded)
Data Precpessing
• Face Detection
• Image Enharenent
• Feature Norimlation
Multi-Modal AI
Feature Extraction
• Face Embedling (ArcFace)
• Gait Recogntion (OpenPose)
Height Estimation
National Missing Database
Matching CCTV Monitoring
Across Public Locations
AI Match Confidence Engine
(Face + Clothing + Gait)
Match
Found?
Geo-Intelligent Alert System
• Notify Nearest Police
• Send SMS to Family
• Generate Location Heatmap
Case Monitoring Dashboard
(Live Tracking & Updates)

1. DATA INGESTION LAYER (Input)
2. MULTI-MODAL AI CORE (Analysis)
3. ORCHETTATION & MATCHING (The Brain)
4. APPLICATION & ACTION LAYER (Output)
PREPORESSING
ESRGAN, Face Alignment
BIOMETRIC ENGINE
(ArcFace)
BEHEVRIOAL ENGINE
(OpenPose, Gait Analysis)
ATTRIBUTE ENGINE
(YOLO, Clothing/Objects)
GEOMETRY ENGINE
Height Estimation
VECTOR DATABASE
(Milvus/Pinecone)
CONFIDENCE ENGINE
THRESHOLD FILTER
ALERT SYSTEM
COMMUNICATION
(SMS/Family
VISUALIZATION
(Live Heatmaps)

Technologies to be used in the solution:

- **1. Artificial Intelligence & Machine Learning**
  - **Face Recognition**
    - **ArcFace / FaceNet / DeepFace**
      - Extracts facial embeddings
      - Performs similarity matching
  - **2. Object & Clothing Detection**
    - **YOLOv8**
      - Detects person, clothing color, accessories
      - Real-time inference
  - **3. Gait Recognition**
    - **OpenPose / MediaPipe Pose**
      - Extracts walking pattern features
      - Identifies individuals even if face is hidden
  - **4.Age Progression Model**
    - **GAN-based Model (StyleGAN / Face Aging GAN)**
      - Predicts how a person may look after years

- **2. Deep Learning Frameworks**
  PyTorch – Model development & training
  TensorFlow / Keras – Deployment & optimization
  OpenCV – Image processing

- **3. Data Processing & Backend**
  Python – Core AI logic
  FastAPI / Flask – Backend API
  Firebase / MongoDB – Database storage
  MySQL / PostgreSQL – Structured data management

- **4. Real-Time Video Processing**
  OpenCV Video Capture
  RTSP Streaming
  NVIDIA CUDA (Optional) – GPU acceleration

◆ **5. Geo-Intelligence & Mapping**
Google Maps API
Leaflet.js
GeoPandas
Used for:
Heatmaps
Location-based alerts
Tracking sightings

◆ **6. Mobile & Web Application**
📱 **Mobile App**
Flutter / React Native
Police & Guardian interface
🌐 **Web Dashboard**
React.js
Node.js
Live monitoring panel

◆ **7. Cloud & Deployment**

- **AWS / Google Cloud / Azure**
- **Docker**
- **CI/CD Pipeline**
- Edge deployment support

◆ **8. Security & Privacy**

- AES Encryption
- Role-based Authentication
- Secure API Endpoints
- GDPR / Data Protection Compliance

Estimated implementation cost (optional):

Component	Estimated Cost (INR)
Cloud GPU (3 months – AWS/GCP)	₹40,000 – ₹60,000
Database & Hosting	₹10,000
Domain & Server	₹5,000
App Development & Testing	₹25,000
Miscellaneous / Deployment	₹10,000

Add as per the requirements for the hackathon:

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Thank You